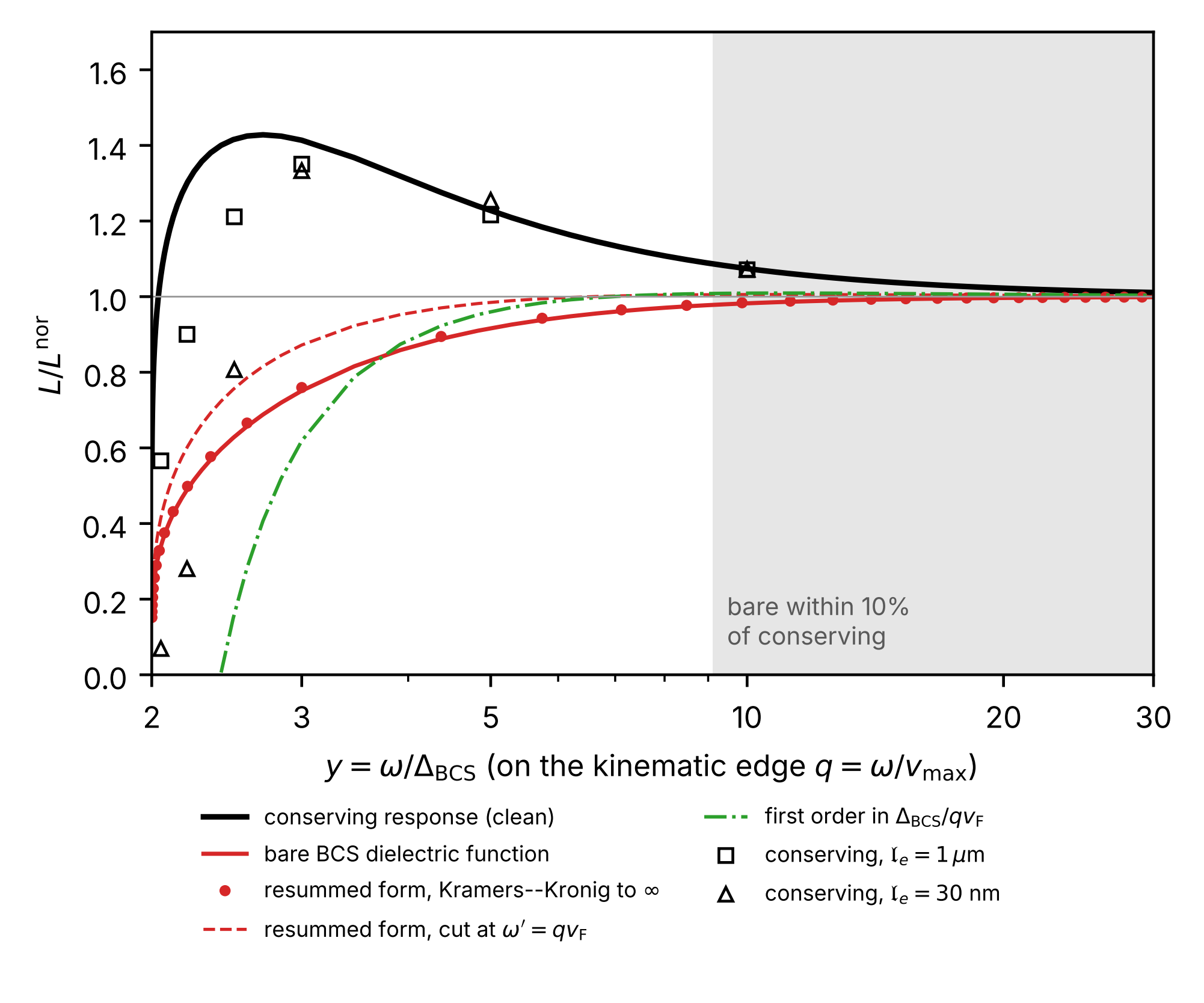
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
L/L^{nor}
2
3
5
10
20
30
$y = \omega/\Delta_{\mathrm{BCS}}$ (on the kinematic edge $q = \omega/v_{\mathrm{max}}$)
bare within 10%
of conserving
conserving response (clean)
bare BCS dielectric function
resummed form, Kramers--Kronig to ∞
resummed form, cut at $\omega' = qv_{\mathrm{F}}$
first order in $\Delta_{\mathrm{BCS}}/qv_{\mathrm{F}}$
conserving, $\ell_e = 1\,\mu$m
conserving, $\ell_e = 30$ nm